

POLITECNICO DI MILANO







Investments evaluation

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Agenda

Investiments

- What is an investment
  Types of investment

Investment evaluation

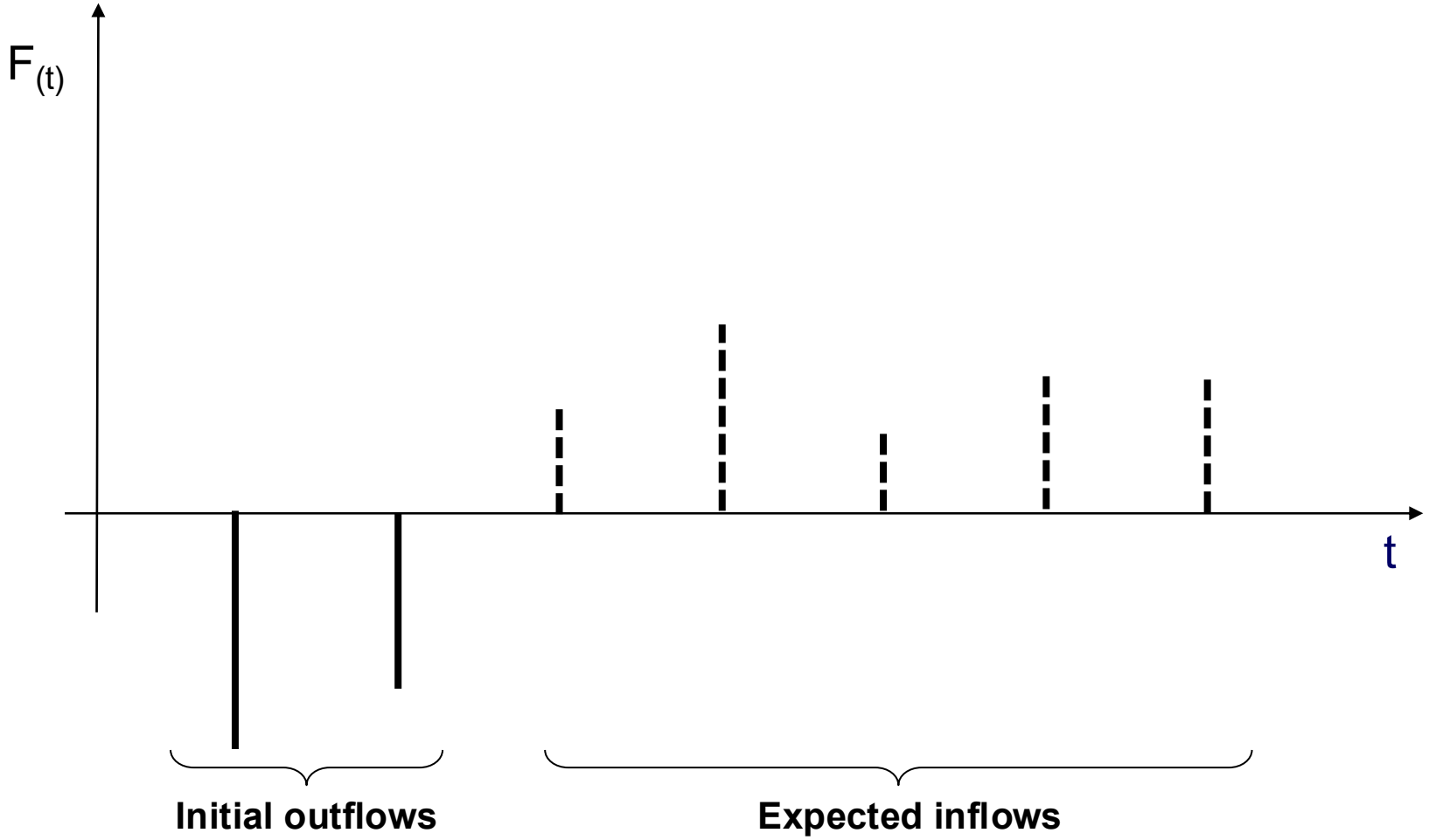
- Phases of analysis
  - Mandatory and non-mandatory investments, decision criteria
  - Deterministic criteria
    - Discounted Cash Flow (NPV, IRR, Payback)
    - Non discounted Cash Flow (Pay back, ROI)

Investments

- ***Investiment*:** use of financial resources with the aim of creating economic value in the medium to long term

- **Characteristics of the investments:**
  - Prevalence of cash outflows in the initial periods
  - Existence of a long-term time horizon (useful life)
  - Future remuneration of the initially invested monetary resources (capital recovery) and return on the invested amount (investment yield)
  - Uncertain outcomes: risk analysis
  - Irreversible decision

Investments



Typologies of the investments

A firm may invest in:

- Real assets (tangible or intangible) aimed at the production of goods and services
- Expansion
- Replacement
- Automation
- Adoption of a new technology
- Introduction of new products or services
- Innovation and R&D
- Financial assets → holding of securities
- Acquisition of equity stakes (shares of another company)

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Investments evaluation

Assessment of the impact that a given investment has on the entity that is adopting such investment (firm, but also individual or public entity)

Objective of the evaluation

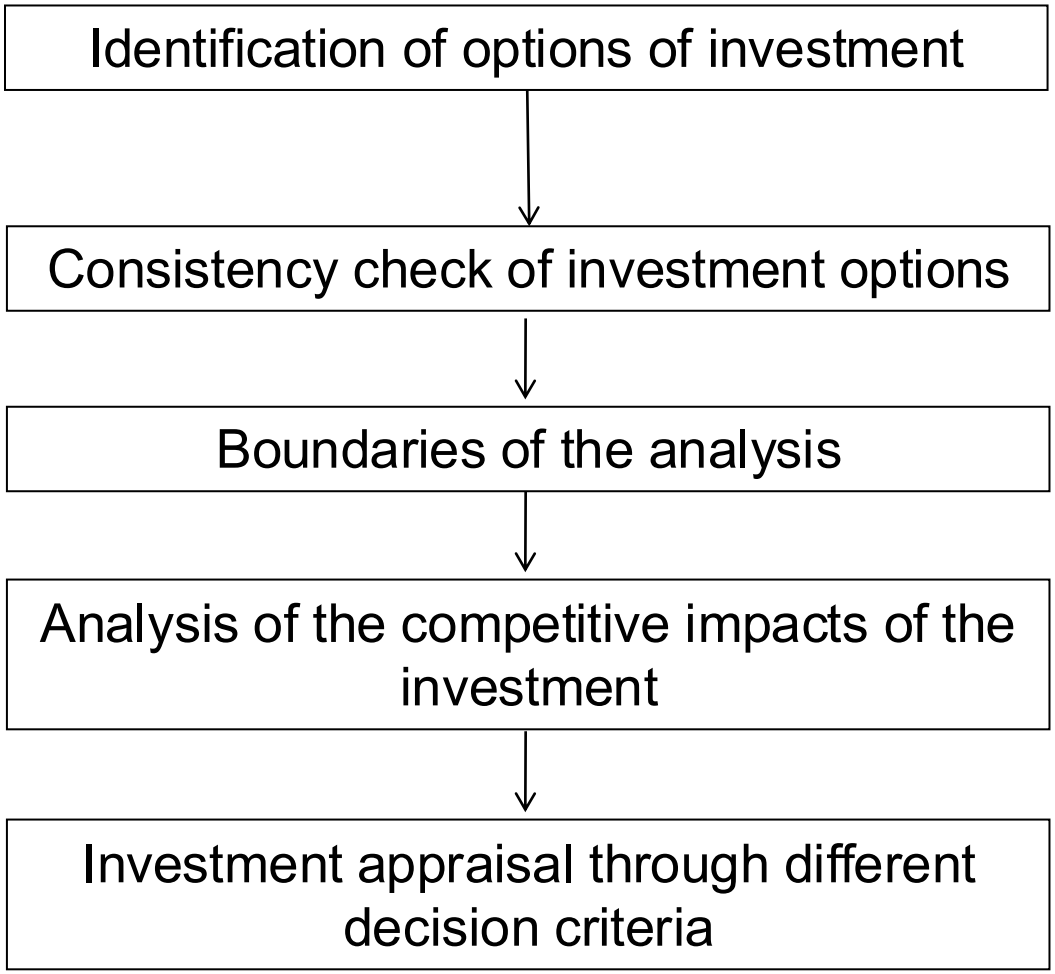
Generate sufficient information to allocate resources:

- only to projects that create value (i.e. where the difference between benefits and costs is positive)
- to projects that create the highest value

Problems

- How to evaluate deferred costs and benefits?
- How to take into account possible budget constraints?

Phases of analysis



Identification of options of investment

There are two typical situations

- Non-mandatory investments: cases in which the firm may or may not undertake the investment
  - Among the alternatives there is the option of not investing, the “base case”
  - The other alternatives are analysed in differential terms with respect to the base case

- Mandatory investments: the firm can at most choose among different investment alternatives
  - “Base case”: any of the investment alternatives

Identification of options of investment

Typical errors in the identification of alternatives are:

- Non-mandatory investments: assuming that in the base case the firm's results remain unchanged

- Mandatory / non-mandatory investments
  - If a non-mandatory investment is considered mandatory → risk of accepting value-destroying investments
  - If a mandatory investment is considered non-mandatory → risk of incurring in business related issues

Consistency of projects

- It is necessary to:
  - Not forget essential “projects” required to implement the overall investment (project bundling)
  - Avoid considering investment portfolios as “integrated” when this is not necessary

Boundaries of the analysis

- By intervening on one part of the company, the functioning of other organisational units is also affected

- Therefore:
  - A process map is required
  - The effect of the investment on activities must be analysed this way:
    - how inputs are modified
    - how outputs are modified
    - What activities are made “redundant” and out of the scope of the analysis

Competitive impacts of investments

- It is necessary to understand how the decision affects the different sources of competitive advantage of the firm a

- Cost differentials at the business level
- Attractiveness differentials at the business level
- Portfolio evaluation differentials

Decision criteria

- ***Decision criteria***
  - Wheter to make or not the investment (*acceptance criteria*)
  - which investment to undertake among mutually exclusive investments (*ranking criteria*)

Decision criteria may be:

- ***Deterministic criteria*: when the following hypothesis hold:**
  - **all investments are characterised by a comparable level of risk**
  - **investments do not change the firm's risk position**

- Pseudo-deterministic approaches
- Stochastic approaches

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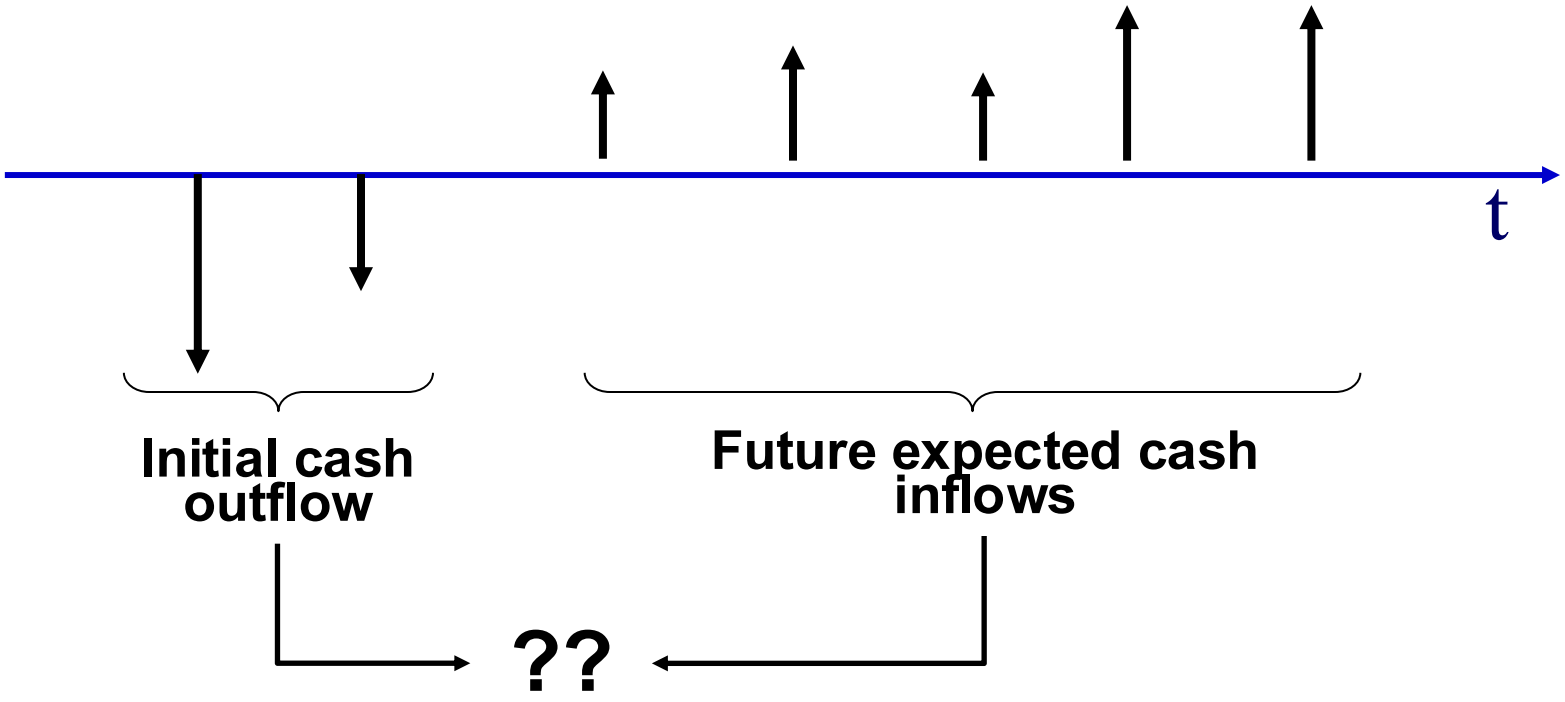
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Deterministic criteria

1. ***Discounted Cash Flow*** (DCF): cash flow distribution over time
    1. Net Present Value (NPV)
    2. Profitability Index (PI)
    3. Internal Rate of Return (IRR)
    4. Discounted Pay back
2. ***Non Discounted Cash Flow*** (Non DCF): do not consider cash flow distribution over time
    1. Pay back
    2. Return On Investments (ROI)

The role of time

- Value creation and absorption are generally distributed over multiple points in time and, above all, over multiple accounting periods



How can cash flows occurring at different points in time be compared?
??

The time value of money

One euro today is worth more than one euro tomorrow

What is the value M of an amount S invested today at an interest rate i for N years?

S = 10.000

- i = 6%
- N = 5

Compound interest

Periods	Amount of money at the beginning of the period	Yield	Total amount	Final amount
1	10.000	600	10.600	
2	10.600	636	11.236	
3	11.236	674	11.910	
4	11.910	714	12.624	
5	12.624	757	13.382	13.382

The time value of money

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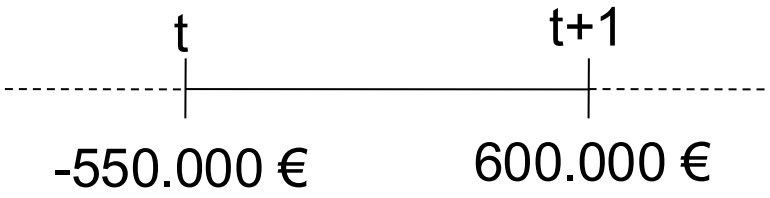
$$M = S(1+i)^N$$ *Capitalization*

$$S = \frac{M}{(1+i)^N}$$ *Discounting*

Discounting: an example

- A firm would like to purchase a property today for €550,000, knowing that it can resell it in one year for €600,000: is it worthwhile?

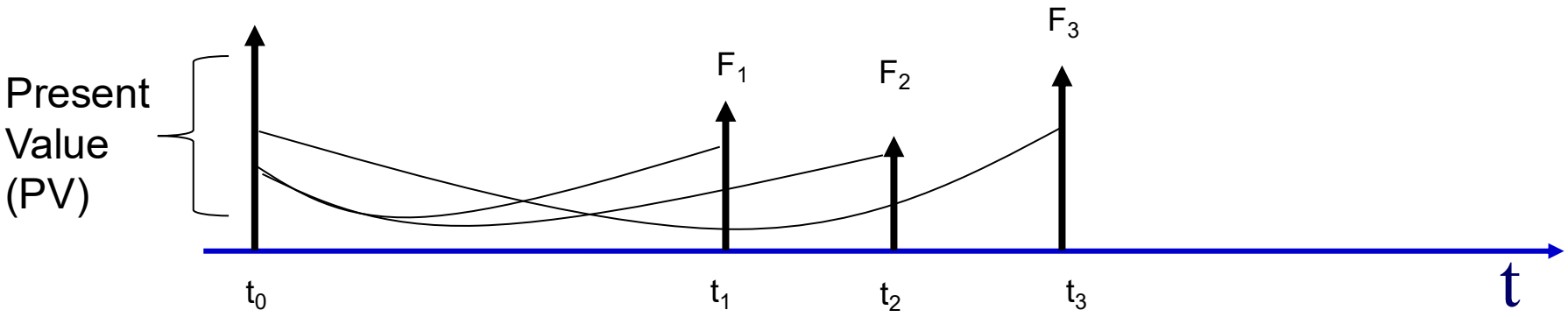
- Simply summing costs and returns would not be correct due to their different timing



- Assume that by investing €550,000 in government bonds the firm would obtain an annual return of 10% (risk-free alternative investment)
- After one year the firm would have: 550.000*(1+0,1) = 605.000 €

  → it is not worthwhile for the firm to purchase the property, but rather to invest in government bonds

The role of time



$$PV = \frac{F_1}{(1+k)} + \frac{F_2}{(1+k)^2} + \cdots + \frac{F_n}{(1+k)^n} = \sum_{t=0}^{n} \frac{F_t}{(1+r)^t}$$

Discounting

- **Discounting:** calculation of the present value of future cash flows taking into account:
  - Growth of prices over time (inflation)
  - Cost of using money for an investment with a certain degree of risk (risk)
  - Cost of debt capital
  - Availability of alternative investments (opportunity cost)

- Through discounting, the Present Value is calculated considering
  - Hurdle rate = minimum required rate of return of the investment
  - Opportunity cost of capital = return forgone or not pursued by investing in the project rather than in a risk-free investment

Reference principles for DCF criteria

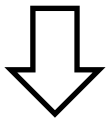
- When applying DCF criteria, two fundamental reference principles must be considered:
  - a financial/cash flow logic is adopted
  - an incremental logic is applied

- In addition, it is necessary to clarify:
  - the financing policy of investment projects (equity / debt)
  - the tax logic considered (before / after taxes)

Financial logic

- Adopting a financial logic means giving relevance not to economic events but to financial ones

- When applying DCF criteria, all and only revenues and costs associated with an actual cash inflow or outflow must be considered



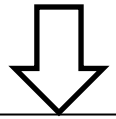
For example, depreciation and amortizations must not be considered, as they do not involve any cash outflow*

*however: they must not be forgotten!

Incremental logic

- Adopting an incremental logic means considering all and only incremental (or differential) cash flows **with respect to the base case**

- Moreover, investment evaluation must ignore all effects of decisions already taken by the firm



Cash flows already committed, or sunk costs, must therefore not be considered

- Similarly, **all and only avoidable costs** must be considered, i.e. those costs that are actually affected by the specific decision being analysed

Criteria DCF: NPV

Net Present Value (NPV)

- Represents the increase in value generated by the investment

NPV = Value generated – Absorbed value

- additional revenues genereated by the investments

- costs of the investments

- Indirect costs and benefits are also included in the appraisal, as well as the impact of the investment on existing activities of the firm
- Costs eliminated thanks to the investment → differential revenues
  Reduction of revenues for other products → differential costs

NPV Calculation

NPV: sum of discounted Net Cash Flows associated with the investment

- $NPV = \sum_{t=0}^{\infty} \frac{NCF(t)}{(1+k)^t}$ or $NPV = \sum_{t=0}^{T} \frac{NCF(t)}{(1+k)^t} + \frac{V_T}{(1+k)^T}$
- If the investment is concentrated at the beginning:

$$NPV = \sum_{t=1}^{T} \frac{CF(t)}{(1+k)^t} + \frac{V_T}{(1+k)^T} - I_0$$

  - $\mathbf{I_0}$: initial financial disbursement
  - **T**: time horizon
  - **k**: discount rate
  - **CF(t)**: differential cash flows related to the investment
  - $\mathbf{V_T}$: residual value (terminal value)

- Acceptance criteria: NPV≥0

  Ranking criteria: preferisco A a B se $NPV_A > NPV_B$

DCF: NPV

- The CEO of a company intends to purchase a new plant and wonders about the convenience of the investment
- The investment has a time horizon of 4 years and is expected to generate the cash flows shown in the table

Year	Cash Flow
0	Disbursement of 1.000.000 € of a new plant
1	Differential revenues of 390.000 €
2	Differential revenues of 325.000 €
3	Differential revenues of 340.000 €
4	Differential revenues of 150.000 €

- The residual value of the plant is expected to be €200,000 (to be sold)
- A discount rate of 10% is used

Solution

NPV =

- $-1{,}000{,}000$
- $390{,}000 / (1 + 0.10)$
- $325{,}000 / (1 + 0.10)^2$
- $340{,}000 / (1 + 0.10)^3$
- $150{,}000 / (1 + 0.10)^4$
- $200{,}000 / (1 + 0.10)^4$

$= -1{,}000{,}000 + 354{,}545 + 268{,}525 + 255{,}447 + 239{,}054 =$ **117,571**

NPV > 0 → it is worthwhile to purchase the plant!

Discount rate k - calculation

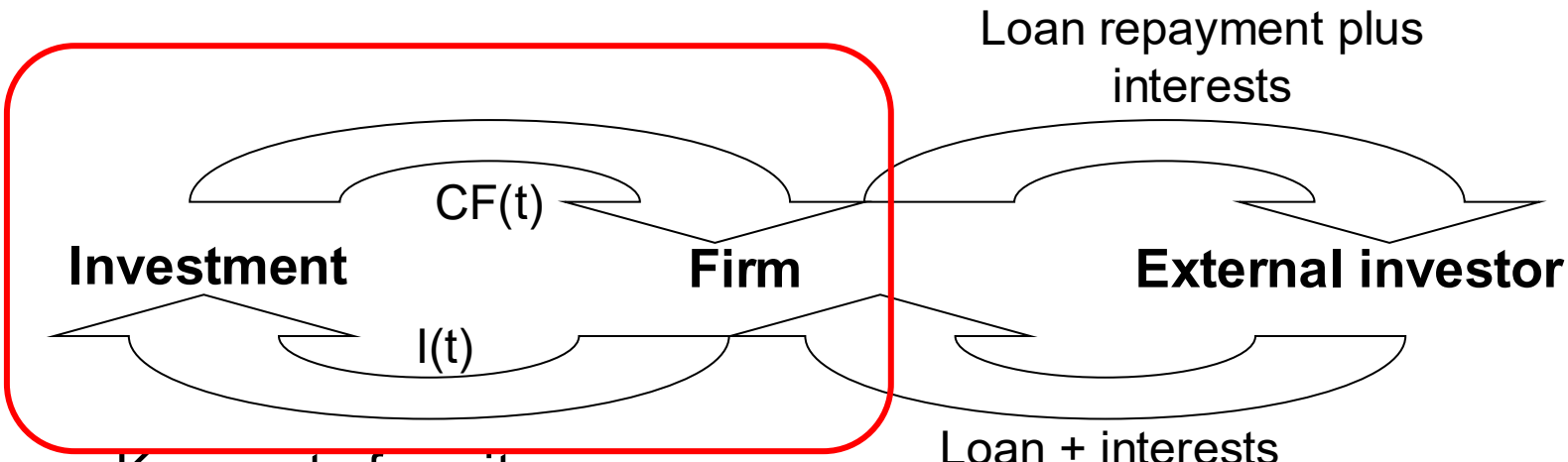
- Financial analysts or financial investors associate a certain level of risk to each firm, depending on:
  - Technologies used
  - Markets served
  - Countries in which the firm operates
  - Financial leverage (D/E)

- As a first approximation, k is firm-specific: each firm evaluates all investments using the same k.
  - k depends on the financial structure of the firm. Two approaches exist:
    - Equity capital approach
    - Invested capital approach

- Creditors require an interest rate i on debt capital.
- Shareholders require a return s on equity.

Discount rate k - calculation

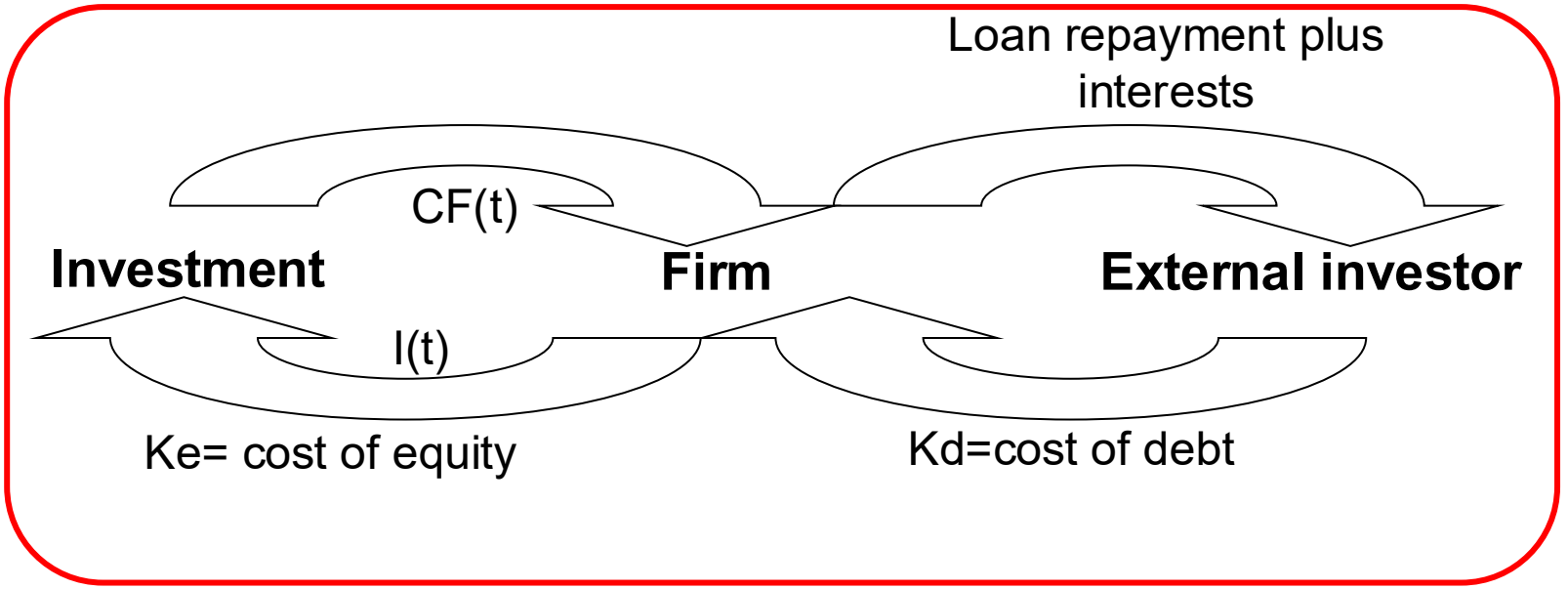
1. ***Equity capital approach:*** *the investment is analysed from the shareholder's perspective. All cash flows between the firm and external investors represent financial flows for shareholders.*



- Ke= cost of equity

Discount rate k - calculation

2. ***Invested capital approach:*** *all actors are considered as part of a single system. Cash flows between the firm and external financiers are not considered. k (or Weigted Average Cost of Capital (WACC)) is calculated as the weighted average of i and s.*



- k (or WACC) is calculated as the weighted average between i and s

$$WACC = k_e \frac{E}{D+E} + k_d(1-T)\frac{D}{D+E}$$

Discount rate - WACC

$$WACC = k_e \frac{E}{D+E} + k_d(1-T)\frac{D}{D+E}$$

The cost of the debt is tax-deductible.
States provide fiscal incentives to raise debt instead of equity.

Cash Flows structuration

YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR …

Differential revenues
- Differential costs
- Interests expense

- Depreciation & Amortizations (D&A)

EBT

or

TAXABLE INCOME

- Tax

Net Income

+ Depreciation & Amortizations (D&A)
- Initial EQUITY investment disbursement on the plant/machine (Year 0)

- Total Debt reimbursement (Final Year)
Selling value after tax: management of the capital gain vs capital losses with respect to taxes

Cash Flow

Discounting with the Cost of Equity

Free Cash Flows to Equity (FCFE)

Cash Flows structuration

YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5

Differential revenues
- Differential costs
- Depreciation & Amortizations (D&A)

EBIT

or

TAXABLE INCOME

- Tax

Net Income

+ Depreciation & Amortizations (D&A)
- Initial investment disbursement on the plant/machine - CAPEX

Selling value after tax: management of the capital gain vs capital losses with respect to taxes

Cash Flow

Discounting with WACC (invested-capital approach)

Free Cash Flow to the Firm (FCFF)

$$SV_{after\ tax} = SV - T(SV - BV)$$

SV – BV = Saving Value – Book Value

SV = sale of the machine/software, ecc.. The value is given by the market

T = taxation: manage the fiscal effect on the difference between SV and the Book Value.

BV = Book Value – the accounting value of an asset recorded on the company's balance sheet at a given point in time. The value of the machine in the balance sheet, given the depreciation plan.

Book Value=Historical Cost−Accumulated Depreciation

D&A

The systematic allocation of the cost of a tangible (depreciation) or of an intangible (amortizations) fixed asset over its useful life (usually expressed in years).

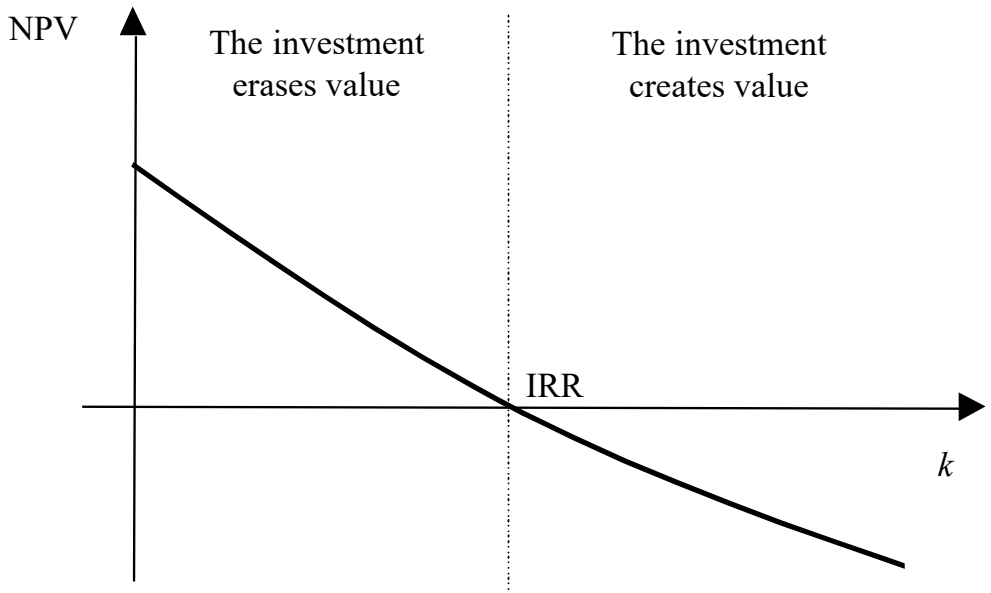
Criteria DCF: IRR

Internal Rate of Return (IRR)

Internal Rate of Return (IRR): discount rate that makes NPV equal to zero.

$$\sum_{t=0}^{T} \frac{NCF(t)}{(1+IRR)^t} = 0$$

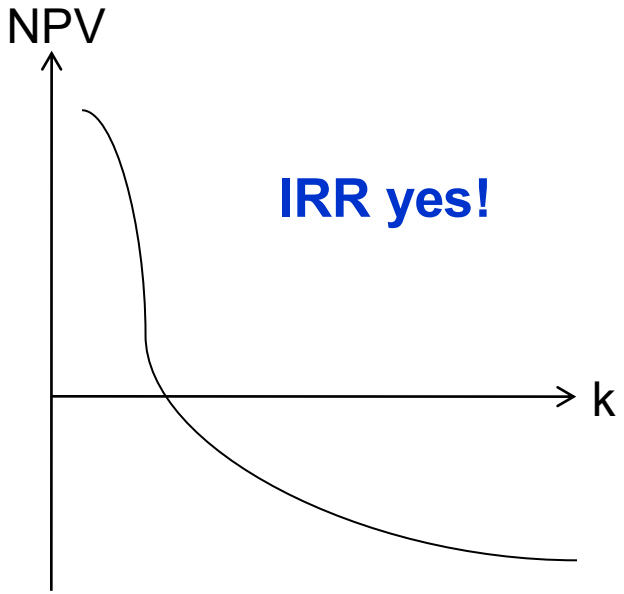
$$IRR = \left(\frac{\text{Exit Value}}{\text{Initial Equity Investment}} \right)^{1/n} - 1$$

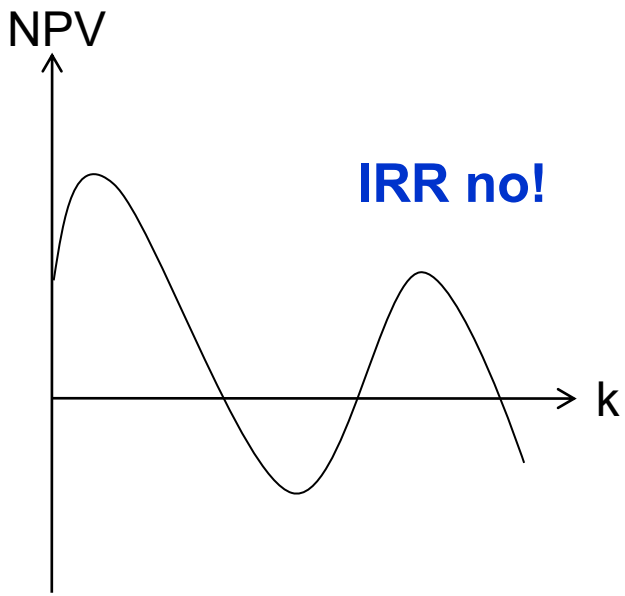


- Acceptance criterion: IRR ≥ k (for example, k may be the WACC).
- If IRR is higher than the cost of capital, the investment is financially attractive.
- Ranking criterion: investment A is preferred to B if IRR_A > IRR_B.

Calculation of the IRR

- Calculating IRR means solving a polynomial of degree T (T equals the investment time horizon). This is not always possible.





- When can the IRR criterio be applied?

Calculate the IRR

- To compute IRR it is not necessary to know the opportunity cost of capital, but it is necessary to apply the criterion.

- IRR may not be unique. When cash flows change sign multiple times, the polynomial has as many roots as sign changes.

- A sufficient condition for a unique IRR is the presence of only one sign change in the cash flows.
  - Initial cash outflows (-) and later on cash inflows (+)
- Returns must cover outflows, otherwise IRR < 0.

Calculate the IRR

$$NPV = -E_0 + \sum_{t=1}^{T} \frac{FCFE_t}{(1+r)^t}$$

In an investment evaluation of an investment fund, r of the investment should be higher than the target IRR of the Fund

E0 = equity invested

FCFE = cash flow for the equity (included exit9

r = required yield (target IRR / cost of equity of the Fund)

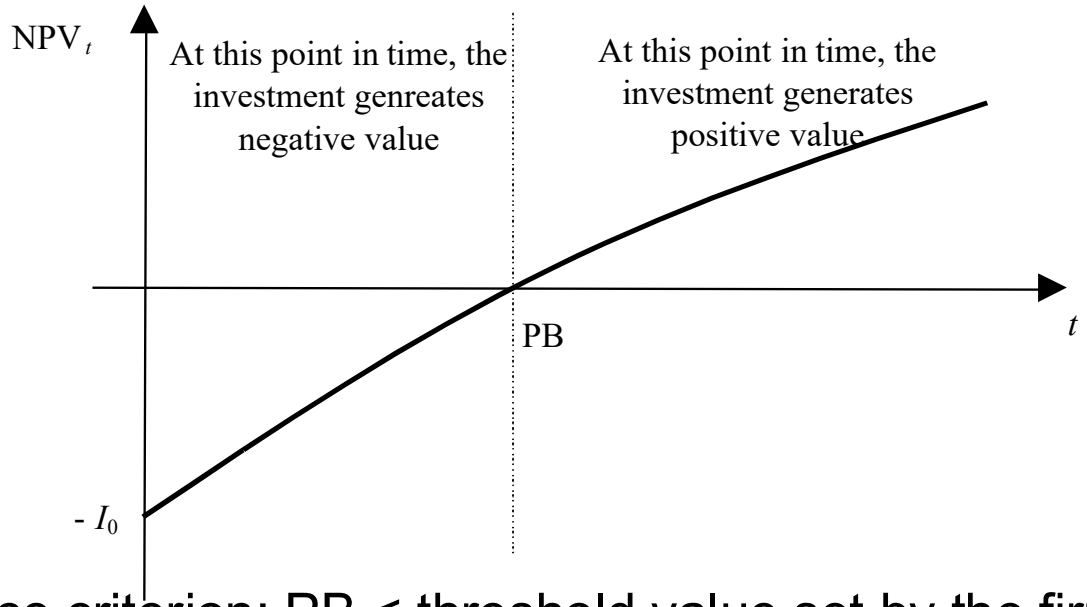
$$0 = -E_0 + \sum_{t=1}^{T} \frac{FCFE_t}{(1+r^*)^t}$$

The IRR is the return generated by the investment.
The NPV measures how much that yield exceeds (or does not reach) the required return.

DCF: discounted pay back period

Discounted payback period (PB): time required for discounted cash flows to recover the initial investment.

$$\sum_{\zeta=0}^{PB} \frac{NCF(\zeta)}{(1+k)^{\zeta}} = 0$$



- Acceptance criterion: PB < threshold value set by the firm.
- Ranking criterion: investment A is preferred to B if PB_A < PB_B

DCF: NPV vs PB
Alternative investments

- Discounted cash flows of two alternative investments:

	0	1	2	3	4
Alternative A	- 2 mln €	2 mln €	0,3 mln €		
Alternative B	- 2 mln €	1 mln €	1 mln €	0,3 mln €	0,3 mln €

- PB_A=1 year
  PB_B=2 year
  Criterion **PB**: better **A** (PB_A<PB_B)

- NPV_A= -2+2+0,3 = 0,3 mln €
  NPV_B= -2+1+1+0,3+0,3 = 0,6 mln €
  Criterion **NPV**: better **B** (NPV_B>NPV_A)

DCF: pay back discounted
Pros and cons of the criterion

- The criterion is conservative.
- It is useful as a complementary indicator to NPV.
  - NPV measures profitability,
  - PB measures liquidity.
- Pros: suitable when future visibility is low.
- Cons: unsuitable for strategic investments with long activation times and for marginal investments.

Non-DCF criteria: payback period

Pay back period (TPB)

Payback period (PB): time when cash flows cover the initial outlay.

$$\sum_{\zeta=0}^{TPB} NCF(\zeta) = 0$$

- Acceptance criterion: TPB < threshold value.
- Ranking criterion: investment A is preferred to B if TPB_A < TPB_B.

- Pros and cons
  - Pros: simplicity.
  - Cons: underestimates the real repayment time and shares the same limits as discounted payback.

Criteri non-DCF: ROI

$$ROI = \frac{\text{EBITDA}}{\text{investment}}$$

- Average operating income can be calculated as:
  EBITDA = revenues generated by the investment minus operating costs, depreciation and amortization.

  Acceptance criterion: ROI > threshold value.
  Ranking criterion: investment A is preferred to B if ROI_A > ROI_B.
- Cons: financial aspects are neglected.
  ROI is a relative criterion and favours smaller investments